

FraudShield: Enterprise Deployment Plan

Phase-by-Phase Production-Grade Implementation & Architectural Blueprint

Project Name:	FraudShield (Tamper-Evident Fraud Prevention Engine)	Target SLA:	99.99% Availability (< 52m downtime/yr)
Team / Org:	Team Stallion — Lloyds Technology Centre	Target TPS:	10,000+ Payment Transactions / sec
Security Model:	DAML Canton Privacy Ledger + ML Risk Engine	Cloud Platform:	Google Cloud Platform (GKE / VPC / KMS)

Executive Summary

FraudShield is an enterprise-grade, privacy-enabled decentralized fraud prevention platform designed to execute multi-node payment verification, Isolation Forest machine learning anomaly scoring, and DAML smart contract hold/consent automation. This document outlines the **5-phase production roll-out strategy** required to transition FraudShield from pilot deployment to full enterprise production.

Phase 1: Infrastructure Landing Zone & High-Availability Core (Weeks 1–2)

Primary Objective: Establish multi-region, zero-trust cloud infrastructure with automated high availability.

- **Multi-Region Cloud Landing Zone:** Deploy GCP Landing Zone using Infrastructure-as-Code (Terraform) across us-central1 and europe-west1 with private Google Access and Cloud Armor WAF.
- **High-Availability MongoDB Cluster:** Provision a 3-node MongoDB Replica Set (Primary/Secondary/Arbiter) with TLS 1.3 in-transit encryption, AES-256 storage encryption, and automated point-in-time recovery (PITR).
- **DAML Canton Distributed Ledger Topology:** Configure enterprise Canton domain with multi-node Sequencer and Mediator nodes using Postgres database persistence and HA consensus consensus synchronization.
- **Hardware Security Module (HSM) Key Management:** Integrate GCP Cloud KMS / HashiCorp Vault to secure party private keys, DAML signing certs, and database encryption keys.

Phase 2: CI/CD Pipeline, Containerization & Immutable Governance (Weeks 3–4)

Primary Objective: Build automated container delivery, code quality gates, and zero-touch artifact pipelines.

- **Enterprise Kubernetes (GKE) Cluster Setup:** Provision GKE Autopilot multi-zone clusters with Workload Identity, Istio Service Mesh, and private cluster endpoints.
- **CI/CD Pipeline Security Gates:** Configure GitHub Actions / GitLab CI with SonarQube static code analysis (>85% coverage gate), Trivy container vulnerability scanning, and OWASP Dependency-Check.

- **DAML DAR Contract Governance:** Automated DAML SDK compilation, unit testing, and DAR artifact uploading to Canton ledger domain with cryptographic release signing.

Phase 3: MLOps Pipeline, Security Controls & Compliance (Weeks 5–6)

Primary Objective: Deploy operational ML model governance, WAF defense, and regulatory audit compliance.

- **Scikit-Learn Isolation Forest MLOps:** Implement MLflow model tracking, automated weekly retraining on incoming ledger feature vectors, model drift detection, and shadow scoring validation.
- **Zero-Trust Security & API Gateway:** Deploy Kong / Apigee API Gateway with OAuth2 / Mutual TLS (mTLS) authentication, rate limiting (100 req/sec/user), and CORS security policy.
- **Regulatory & Compliance Alignment:** Enforce GDPR data anonymization (zero PII stored on immutable ledger), SOC2 Type II audit logging, and ISO 27001 ISMS standard compliance.

Phase 4: Staging Mirror, Load Testing & Chaos Engineering (Weeks 7–8)

Primary Objective: Validate peak throughput performance, ledger resilience, and penetration security.

- **Load & Stress Testing:** Execute k6 / Locust benchmark suites simulating 10,000+ payment transactions per second (TPS) under peak traffic conditions.
- **Chaos Mesh Resilience Testing:** Inject simulated Canton node disconnections, database failovers, and latency spikes to verify automatic consensus recovery.
- **Third-Party Security Audit:** Conduct external CREST-certified penetration testing and DAML smart contract security verification.

Phase 5: Production Rollout, Zero-Downtime Migration & Observability (Weeks 9–10)

Primary Objective: Execute zero-downtime blue-green release, live monitoring, and SRE incident response.

- **Blue-Green / Canary Traffic Cutover:** Deploy new releases via Istio traffic splitting (5% -> 25% -> 100%) with automated health check rollback triggers.
- **Full Observability Stack:** Prometheus metric collection, Grafana real-time operational dashboards, OpenTelemetry distributed tracing, and PagerDuty SRE alerts.
- **24/7 SRE Operational Runbook:** Establish Tier-1 to Tier-3 incident escalation, < 5 minute Mean Time to Detect (MTTD), and < 15 minute Mean Time to Resolve (MTTR).

Production Rollout Summary Matrix

Phase	Key Deliverable	Duration	Target Gate
Phase 1	GCP Landing Zone, MongoDB HA, Canton HA	Weeks 1–2	Multi-node HA Verified
Phase 2	GKE Cluster, CI/CD Pipeline, Trivy Scan	Weeks 3–4	Zero High/Critical Vulnerabilities
Phase 3	MLOps Pipeline, API Gateway mTLS, SOC2	Weeks 5–6	Model Drift < 2% Variance
Phase 4	k6 10k TPS Load Test, Chaos Test, Pen Test	Weeks 7–8	10,000 TPS @ < 120ms Latency
Phase 5	Canary Cutover, Prometheus, 24/7 SRE	Weeks 9–10	99.99% Availability SLA